

# Exercise 3 – Classification I

## Introduction to Machine Learning

### Exercise 1: Logistic regression – softmax link and decision boundary

The whole exercise is only for lecture group A!

#### Learning goals

1. Solve “show equivalence”-type questions
2. Derive the decision boundary of the logistic classifier

Binary logistic regression is a special case of multiclass logistic, or softmax, regression. The softmax function is the multiclass analogue to the logistic function, transforming scores  $\theta^\top \mathbf{x}$  to values in the range  $[0, 1]$  that sum to one. The softmax function is defined as:

$$\pi_k(\mathbf{x}|\theta) = \frac{\exp(\theta_k^\top \mathbf{x})}{\sum_{j=1}^g \exp(\theta_j^\top \mathbf{x})}, k \in \{1, \dots, g\}$$

- a) Show that logistic and softmax regression are equivalent for  $g = 2$ .

#### Solution

As we would expect, the two formulations are equivalent (up to reparameterization). In order to see this, consider the softmax function components for both classes:

$$\pi_1(\mathbf{x}|\theta) = \frac{\exp(\theta_1^\top \mathbf{x})}{\exp(\theta_1^\top \mathbf{x}) + \exp(\theta_2^\top \mathbf{x})}$$

$$\pi_2(\mathbf{x}|\theta) = \frac{\exp(\theta_2^\top \mathbf{x})}{\exp(\theta_1^\top \mathbf{x}) + \exp(\theta_2^\top \mathbf{x})}$$

Since we know that  $\pi_1(\mathbf{x}|\theta) + \pi_2(\mathbf{x}|\theta) = 1$ , it is sufficient to compute one of the two scoring functions. Let's pick  $\pi_1(\mathbf{x}|\theta)$  and relate it to the logistic function:

$$\pi_1(\mathbf{x}|\theta) = \frac{1}{1 + \exp(\theta_2^\top \mathbf{x} - \theta_1^\top \mathbf{x})} = \frac{1}{1 + \exp(-\theta^\top \mathbf{x})}$$

where  $\theta := \theta_1 - \theta_2$ . Thus, we obtain the binary-case logistic function, reflecting that we only need one scoring function (and thus one set of parameters  $\theta$  rather than two  $\theta_1, \theta_2$ ).

- b) Recall that in the binary case we predict  $\hat{y} = 1$  iff  $\pi(\mathbf{x}) \geq \alpha$  for a threshold  $\alpha \in (0, 1)$ . Show that the resulting decision boundary is a (linear) hyperplane.

*Hint*

Derive the value of  $\theta^\top \mathbf{x}$  (depending on  $\alpha$ ) starting from which you predict  $\hat{y} = 1$  rather than  $\hat{y} = 0$ .

### Solution

We evaluate

$$\begin{aligned} \pi(\mathbf{x}) &= \frac{1}{1 + \exp(-\theta^\top \mathbf{x})} = \alpha \\ \Leftrightarrow 1 + \exp(-\theta^\top \mathbf{x}) &= \frac{1}{\alpha} \\ \Leftrightarrow \exp(-\theta^\top \mathbf{x}) &= \frac{1}{\alpha} - 1 \\ \Leftrightarrow -\theta^\top \mathbf{x} &= \log\left(\frac{1}{\alpha} - 1\right) \\ \Leftrightarrow \theta^\top \mathbf{x} &= -\log\left(\frac{1}{\alpha} - 1\right). \end{aligned}$$

$\theta^\top \mathbf{x} = -\log\left(\frac{1}{\alpha} - 1\right)$  is the equation of a linear hyperplane, consisting of all points  $\mathbf{x}$  whose score  $\theta^\top \mathbf{x}$  equals  $-\log\left(\frac{1}{\alpha} - 1\right)$ . Since  $\pi(\mathbf{x})$  is increasing in  $\theta^\top \mathbf{x}$ , we set  $\hat{y} = 1$  for all points on or above this hyperplane, i.e. with  $\theta^\top \mathbf{x} \geq -\log\left(\frac{1}{\alpha} - 1\right)$ .

## Exercise 2: Hyperplanes

### Learning goals

1. Understand that hyperplanes bisect the space with a linear boundary
2. Get a feeling for coefficients in hyperplane equations
3. See how the classification threshold  $\alpha$  shifts the logistic decision boundary

Linear classifiers like logistic regression learn a decision boundary that takes the form of a (linear) hyperplane. Hyperplanes are defined by equations  $\theta^\top \mathbf{x} = b$  with coefficients  $\theta$  and a scalar  $b \in \mathbb{R}$ .

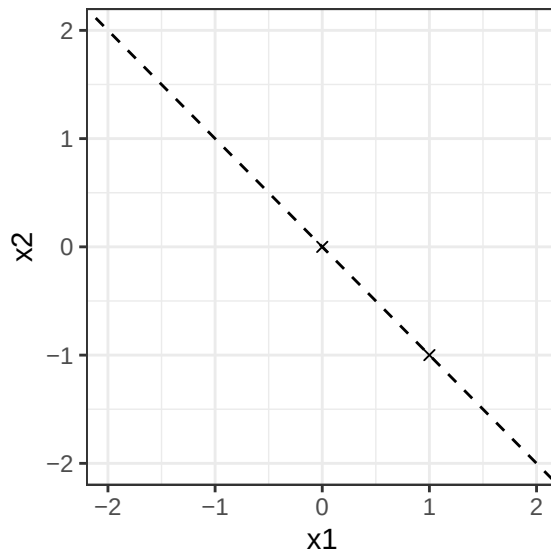
a) In order to see that such expressions actually describe hyperplanes, consider  $\theta^\top \mathbf{x} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$ . Sketch the hyperplanes given by the following coefficients and explain the difference between the parameterizations:

- $\theta_0 = 0, \theta_1 = \theta_2 = 1$
- $\theta_0 = 1, \theta_1 = \theta_2 = 1$
- $\theta_0 = 0, \theta_1 = 1, \theta_2 = 2$

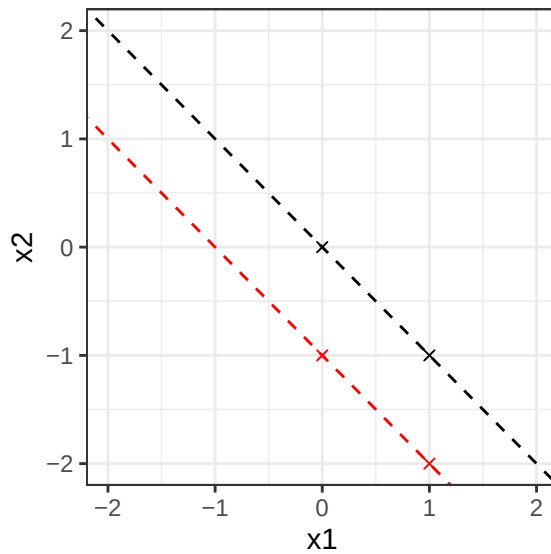
### Solution

A hyperplane in 2D is just a line. We know that two points are sufficient to describe a line, so all we need to do is pick two points fulfilling the hyperplane equation.

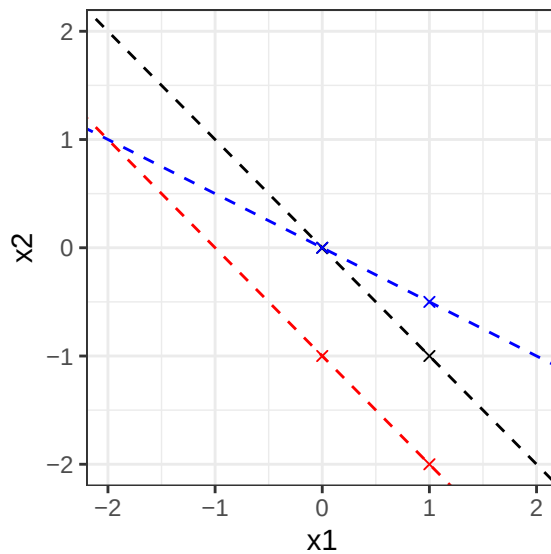
- $\theta_0 = 0, \theta_1 = \theta_2 = 1 \leadsto$  e.g.,  $(0, 0)$  and  $(1, -1)$ . Sketch it:



- $\theta_0 = 1, \theta_1 = \theta_2 = 1 \rightsquigarrow$  e.g.,  $(0, -1)$  and  $(1, -2)$ . The change in  $\theta_0$  promotes a horizontal shift:



- $\theta_0 = 0, \theta_1 = 1, \theta_2 = 2 \rightsquigarrow$  e.g.,  $(0, 0)$  and  $(1, -0.5)$ . The change in  $\theta_2$  pivots the line around the intercept:



We see that a hyperplane is defined by the points that lie directly on it and thus fulfill the hyperplane equation.

- b) Beyond its orientation, a linear classifier also has an *offset* that determines where exactly the boundary sits. Recall that a logistic regression model estimates  $\pi(\mathbf{x}) = \frac{1}{1 + \exp(-\theta^\top \mathbf{x})}$  and predicts  $\hat{y} = 1$  iff  $\pi(\mathbf{x}) \geq \alpha$  for a threshold  $\alpha \in (0, 1)$ . Its decision boundary is

therefore the hyperplane

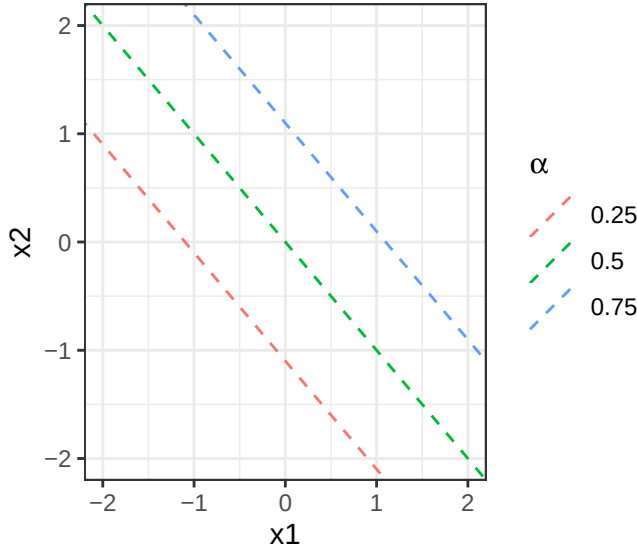
$$\theta^\top \mathbf{x} = -\log\left(\frac{1}{\alpha} - 1\right) =: b(\alpha).$$

Consider the fixed model  $\theta = (\theta_0, \theta_1, \theta_2)^\top = (0, 1, 1)^\top$ . Compute the offset  $b(\alpha)$  for  $\alpha \in \{0.25, 0.5, 0.75\}$ , sketch the three resulting decision boundaries in a single  $(x_1, x_2)$  plot, and describe how the boundary – and the region predicted as class 1 – changes as  $\alpha$  grows. Does the orientation of the boundary change?

### Solution

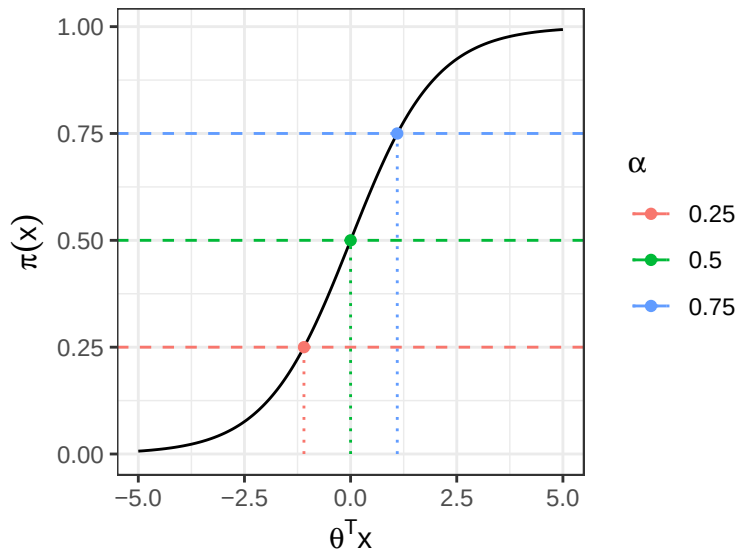
For this model,  $\theta^\top \mathbf{x} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = x_1 + x_2$ , so every decision boundary is a line  $x_1 + x_2 = b(\alpha)$  with slope  $-1$  and intercept  $b(\alpha)$ :

- $\alpha = 0.25$ :  $b(\alpha) = -\log\left(\frac{1}{0.25} - 1\right) = -\log 3 \approx -1.10$
- $\alpha = 0.5$ :  $b(\alpha) = -\log\left(\frac{1}{0.5} - 1\right) = -\log 1 = 0$
- $\alpha = 0.75$ :  $b(\alpha) = -\log\left(\frac{1}{0.75} - 1\right) = -\log \frac{1}{3} = \log 3 \approx 1.10$



As  $\alpha$  increases, the offset  $b(\alpha)$  increases and the boundary shifts *parallel* toward the upper right (larger  $\theta^\top \mathbf{x}$ ). Since we predict  $\hat{y} = 1$  wherever  $\theta^\top \mathbf{x} \geq b(\alpha)$ , the region classified as class 1 (the upper-right halfplane) shrinks: a higher threshold makes the model more conservative about predicting class 1. The orientation does **not** change – the normal vector  $\theta = (1, 1)^\top$  stays fixed, so all three boundaries are parallel. This mirrors part a), where changing the intercept  $\theta_0$  produced a parallel shift of the hyperplane.

Crucially, the fitted model itself – the logistic (S-)curve  $\pi(\mathbf{x}) = \frac{1}{1 + \exp(-\theta^\top \mathbf{x})}$  – does **not** change when we vary  $\alpha$ ; it depends on  $\theta$  alone. Varying  $\alpha$  only moves the *horizontal* cut-off through this fixed curve:



We read off the score  $b(\alpha)$  at which the curve reaches height  $\pi(\mathbf{x}) = \alpha$  (dashed lines) and classify every point with  $\theta^\top \mathbf{x} \geq b(\alpha)$  as class 1. Thus  $\alpha$  is a *post-hoc decision threshold* applied to fixed predicted probabilities, not a re-fitting of the model. The three cut-off scores  $b(\alpha) \in \{-1.10, 0, 1.10\}$  (dotted verticals) are exactly the intercepts of the three parallel decision boundaries in the  $(x_1, x_2)$  plot above.

### Exercise 3: Decision Boundaries & Thresholds in Logistic Regression

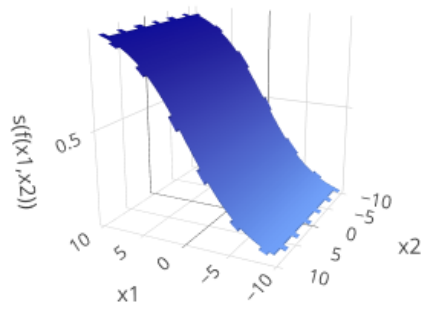
#### Learning goals

1. Understand that logistic regression finds a linear decision boundary
2. Get a feeling for how the parameters  $\theta$  shape predicted probabilities and model confidence
3. Understand the role of the decision threshold  $\alpha$

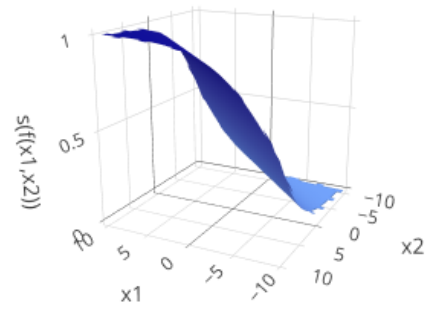
Recall from Exercise 2 that a logistic regression model estimates the probability  $p(y = 1|\mathbf{x}, \theta) = \pi(\mathbf{x}|\theta)$  and assigns  $\hat{y} = 1$  iff  $\pi(\mathbf{x}|\theta) \geq \alpha$  for a threshold  $\alpha \in (0, 1)$ . Whereas Exercise 2 b) varied the threshold for a fixed model, we now study how the parameters  $\theta$  themselves shape the predicted probabilities.

- a) Below you see the logistic function for a binary classification problem with two input features for different values  $\theta^\top = (\theta_1, \theta_2)^\top$  (plots 1-3) as well as  $\alpha$  (plot 4). What can you deduce for the values of  $\theta_1$ ,  $\theta_2$ , and  $\alpha$ ? What are the implications for classification in the different scenarios? In particular, which of the plots (1)–(3) shows the most confident model?

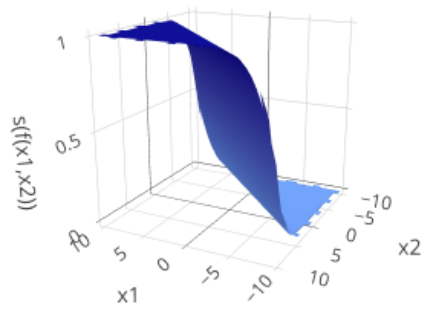
Plot (1)



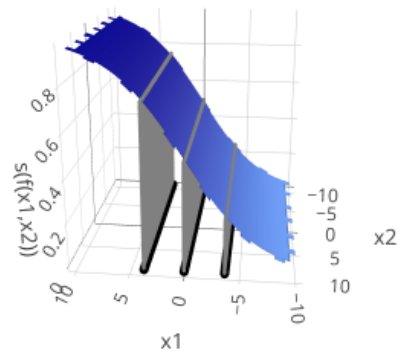
Plot (2)



Plot (3)



Plot (4)



### Solution

We observe

- in plot (1): the logistic function runs parallel to the  $x_2$  axis, so it is the same for every value of  $x_2$ . In other words,  $x_2$  does not contribute anything to the class discrimination and its associated parameter  $\theta_2$  is equal to 0.
- in plot (2): both dimensions affect the logistic function – to equal degree in this case, meaning  $x_1$  and  $x_2$  are equally important. If  $\theta_1$  were larger than  $\theta_2$  or vice versa the hypersurface would be more tilted towards the respective axis. Furthermore, due to  $\theta_1$  and  $\theta_2$  being positive,  $\pi(\mathbf{x})$  increases with higher values for  $x_1$  and  $x_2$ .
- in plot (3): this is the same situation as in plot (2) but the logistic function is steeper, which is due to  $\theta_1, \theta_2$  having larger absolute values. We therefore get a sharper separation between classes (fewer predicted probability values close to 0.5, so we are overall more confident in our decision). As in plot (2), the increasing probability of  $\hat{y} = 1$  for higher values of  $x_1$  and  $x_2$  indicates positive values for  $\theta_1$  and  $\theta_2$ .
- in plot (4): this is the same situation as in plot (1). The different values for  $\alpha$  represent different thresholds: a high value (leftmost line) means we only assign class 1 if the estimated class-1 probability is large. Conversely, a low value (rightmost line) signifies we are ready to predict class 1 at a low threshold – in effect, this is the same as the previous scenario, only the class labels are flipped. The mid line corresponds to the common case  $\alpha = 0.5$  where we assign class 1 as soon as the predicted probability is more than 50%.

In summary, **plot (3)** shows the most confident model: its steeper logistic surface pushes predicted probabilities closer to 0 or 1, yielding the sharpest class separation.

- b) In Exercise 2 b) we simply read off the offset  $b(0.5) = 0$  for the common threshold  $\alpha = 0.5$ . Derive this special case directly, starting from  $\pi(\mathbf{x}) = 0.5$ .

### Solution

At the decision boundary the model is exactly indifferent between the two classes, i.e.,  $\pi(\mathbf{x}) = \alpha = 0.5$ :

$$\begin{aligned}\pi(\mathbf{x}) = \frac{1}{1 + \exp(-\theta^\top \mathbf{x})} = 0.5 &\Leftrightarrow 1 + \exp(-\theta^\top \mathbf{x}) = 2 \\ &\Leftrightarrow \exp(-\theta^\top \mathbf{x}) = 1 \\ &\Leftrightarrow \theta^\top \mathbf{x} = 0.\end{aligned}$$



The 0.5 threshold therefore leads to the coordinate hyperplane  $\theta^\top \mathbf{x} = 0$  and divides the input space into the positive “1” halfspace where  $\theta^\top \mathbf{x} \geq 0$  and the “0” halfspace where  $\theta^\top \mathbf{x} < 0$ .

- c) Explain when it might be sensible to set  $\alpha$  to 0.5.

### Solution

When the threshold  $\alpha = 0.5$  is chosen, the losses of misclassified observations, i.e.,  $L(\hat{y} = 0 \mid y = 1)$  and  $L(\hat{y} = 1 \mid y = 0)$ , are treated equally, which is often the intuitive thing to do. It means  $\alpha = 0.5$  is a sensible threshold if we do not wish to avoid one type of misclassification more than the other. If, however, we need to be cautious to only predict class 1 if we are very confident (for example, when the decision triggers a costly therapy), it would make sense to set the threshold considerably higher.

## Exercise 4: Thresholds and predictions in practice

The whole exercise is only for lecture group B!

### Learning goals

1. Compute predictions of a logistic model by hand
2. Understand how the threshold  $\alpha$  trades off the two error types

Consider a fitted binary logistic regression model with parameter vector  $\theta = (\theta_0, \theta_1, \theta_2)^\top = (-1, 2, 1)^\top$ , so that

$$\pi(\mathbf{x}) = \frac{1}{1 + \exp(-(\theta_0 + \theta_1 x_1 + \theta_2 x_2))}.$$

Recall that we predict  $\hat{y} = 1$  iff  $\pi(\mathbf{x}) \geq \alpha$ .

- a) Compute the predicted probability  $\pi(\mathbf{x})$  for the following observations and classify them using the default threshold  $\alpha = 0.5$ :

- $\mathbf{x}^{(1)} = (x_1, x_2)^\top = (1, 1)^\top$
- $\mathbf{x}^{(2)} = (1, -1)^\top$
- $\mathbf{x}^{(3)} = (0, 0)^\top$

### Solution

We plug each observation into  $\theta^\top \mathbf{x} = \theta_0 + \theta_1 x_1 + \theta_2 x_2$  and apply the logistic function:

- $\mathbf{x}^{(1)}$ :  $\theta^\top \mathbf{x} = -1 + 2 \cdot 1 + 1 \cdot 1 = 2 \Rightarrow \pi(\mathbf{x}) = \frac{1}{1 + \exp(-2)} \approx 0.88 \Rightarrow \hat{y} = 1$

- $\mathbf{x}^{(2)}$ :  $\theta^\top \mathbf{x} = -1 + 2 \cdot 1 + 1 \cdot (-1) = 0 \Rightarrow \pi(\mathbf{x}) = \frac{1}{1+\exp(0)} = 0.5 \Rightarrow \hat{y} = 1$  (exactly on the boundary, where  $\pi(\mathbf{x}) = \alpha$ )
- $\mathbf{x}^{(3)}$ :  $\theta^\top \mathbf{x} = -1 + 2 \cdot 0 + 1 \cdot 0 = -1 \Rightarrow \pi(\mathbf{x}) = \frac{1}{1+\exp(1)} \approx 0.27 \Rightarrow \hat{y} = 0$

Note that  $\mathbf{x}^{(2)}$  lies exactly on the  $\alpha = 0.5$  decision boundary  $\theta^\top \mathbf{x} = 0$  derived in Exercise 3 b).

- b) For observation  $\mathbf{x}^{(1)}$ , from which threshold  $\alpha$  onwards would the model predict  $\hat{y} = 0$  instead?

#### Solution

Since  $\hat{y} = 1$  iff  $\pi(\mathbf{x}) \geq \alpha$ , the prediction for  $\mathbf{x}^{(1)}$  switches to  $\hat{y} = 0$  as soon as  $\alpha > \pi(\mathbf{x}^{(1)}) \approx 0.88$ . In other words, we would only predict class 0 for this observation if we demanded a class-1 probability of more than about 88%.

- c) The model is used in a hospital to decide whether to start an expensive and invasive therapy ( $\hat{y} = 1$ ). Would you rather choose a high or a low threshold  $\alpha$ ? Which type of error (false positive / false negative) does your choice avoid, and at what cost?

#### Solution

If the therapy itself is costly and invasive, we want to be sure a patient really needs it before starting it, i.e., only predict  $\hat{y} = 1$  when we are very confident. This calls for a **high** threshold  $\alpha$ : it reduces **false positives** (healthy patients undergoing an unnecessary, harmful therapy), at the price of more **false negatives** (patients who would benefit are not treated).

The “right” choice therefore depends on the relative costs of the two error types. If missing a case were far more dangerous than an unnecessary treatment (e.g., a life-threatening but easily treatable disease), we would instead prefer a **low**  $\alpha$  to avoid false negatives.